

Simple And Low-Cost GPS CLOCK

FAYAZ HASSAN

R real-time clock (RTC) integrated circuits (ICs) are used for reasonably accurate time displays. The accuracy of RTCs mostly depends on the crystal. In a long run, a small-time deviation will be observed in the RTC circuit due to seasonal and atmospheric temperature changes. In case of a Global Positioning System (GPS) clock, the time/time stamp is received from the satellite(s), and is highly accurate. This project is based on a GPS clock using AVR ATmega8A microcontroller (MCU).

Circuit and working

Circuit diagram of the simple and low-cost GPS clock is shown in Fig. 1. It is built around ATmega8A (IC1), NEO 6M GPS receiver module (GPS1) along with an antenna, 16x1 LCD (LCD1) and a few other components.

NEO 6M GPS receiver module
receives National Marine Electron-

ics Association (NMEA) data continuously and transfers the same to ATmega8A MCU. The MCU processes the data and picks the date stamp from the received data string. The following data format is searched in the received data, which is used to pick the time stamp:

\$GPGGA, 143621,

The MCU checks for the keyword \$GPGGA and reads the next six characters of the received string and displays it on LCD1. This process repeats in a cyclic process, and the latest time is displayed.

The time stamp received is always in Universal Time Coordinated (UTC) format, which matches with Greenwich Mean Time (GMT). To convert it to local time, add proper time constant to UTC time, which will depend on the country and city. In the program, the local time constant for Indian time is set to + 5:30, but you

can modify it to set a different value.

The local time constant is declared in C program as:

```
int ADJ[2] = {5,30};
```

Initial setup

After connecting the circuit, upload GPSclock.hex file to ATmega8A MCU using any AVR programmer through ISP port. Once the program is loaded, blinker LED1 will glow for one second followed by 'NEO 6M GPS Clock' message on LCD1. By default, it will display in UTC format.

To change to local time constant, press any switch (S2 or S3) to display the existing value. By pressing hours and minutes switch, the local time constant changes. If no switch is pressed for more than four seconds, the value is saved in EEPROM of the MCU.

The GPS module will initialise and try to connect to the satellite for data

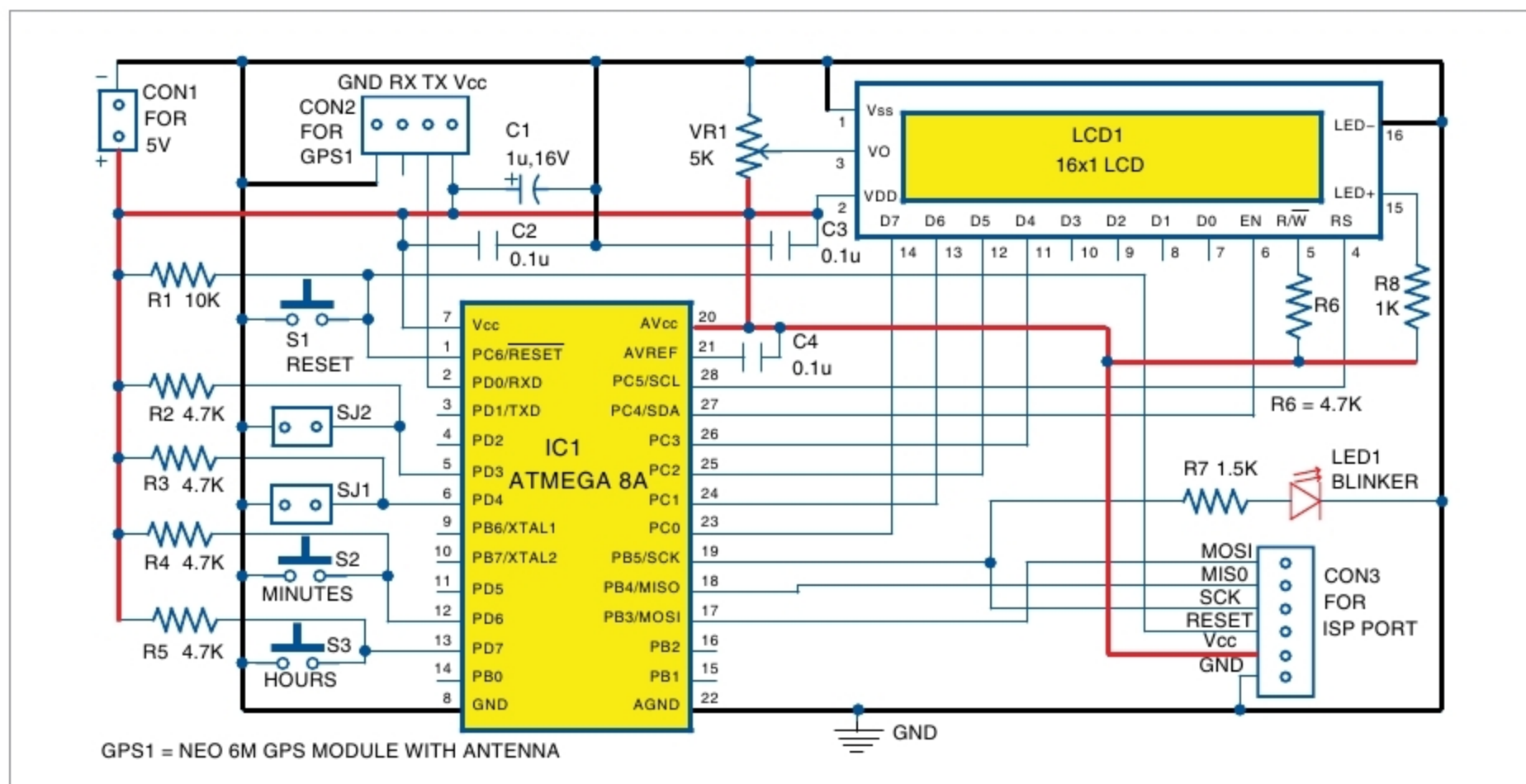


Fig. 1: Circuit diagram of GPS clock